Indian Institute of Technology Guwahati
Department of Electronics and Electrical Engineering
EE101 - Basic Electronics

Tutorial - 9 **Date: Oct 19, 2023**

1. For the emitter-bias network in the active region (base-emitter forward bias, base-collector reverse bias) Fig.1, let $V_{CC} = +18V,\ \ \beta = 60,\ R_B = 430k\Omega,\ \ R_C = 2k\Omega,\ R_E = 1k\Omega$. Determine the following (assume $V_{BE} = 0.7\,V$):
   (a) I_B
   (b) I_C
   (c) V_{CE}
   (d) V_C
   (e) V_E
   (f) V_B
   (g) V_{BC}

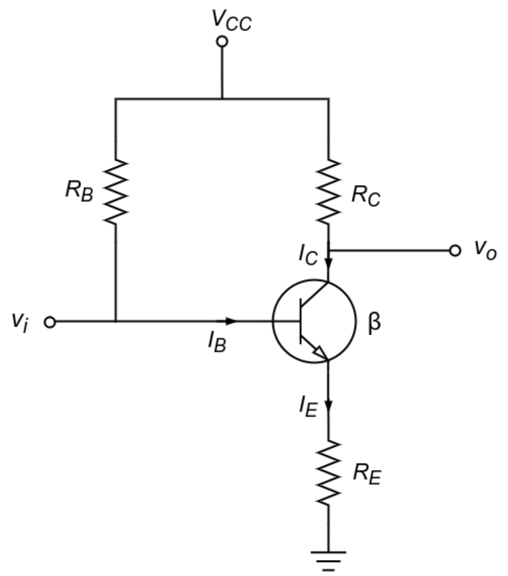

Figure 1

Solution :

(a)

$$\begin{aligned}
I_B &= \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E} \\
&= \frac{18 - 0.7}{430k\Omega + (61)(1k\Omega)} \\
&= \frac{17.3}{491} \\
&= 35.23\mu A
\end{aligned}$$

(b)

$$\begin{aligned}
I_C &= \beta I_B \\
&= (60)(35.23\mu A) \\
&= 2.11mA
\end{aligned}$$

(c)

$$\begin{aligned} I_E &= (\beta + 1)I_B \\ &= (61)(35.23\mu A) \\ &= 2.15mA \end{aligned}$$

$$\begin{aligned} V_{CE} &= V_{CC} - I_C R_C - I_E R_E \\ &= 18V - (2.11mA * 2k\Omega) - (2.15mA * 1k\Omega) \\ &= 18V - 6.37V \\ &= 11.63V \end{aligned}$$

(d)

$$\begin{aligned} V_C &= V_{CC} - I_C R_C \\ &= 18V - (2.11mA)(2k\Omega) \\ &= 18V - 4.22V \\ &= 13.78V \end{aligned}$$

(e)

$$\begin{aligned} V_E &= V_C - V_{CE} \\ &= 13.78V - 11.63V \\ &= 2.15V \end{aligned}$$

or

$$\begin{aligned} V_E &= I_E R_E \\ &= (2.15mA)(1k\Omega) \\ &= 2.15V \end{aligned}$$

(f)

$$\begin{aligned} V_B &= V_{BE} + V_E \\ &= 0.7V + 2.15V \\ &= 2.85V \end{aligned}$$

(g)

$$\begin{aligned} V_{BC} &= V_B - V_C \\ &= 2.85V - 13.78V \\ &= -10.93V \end{aligned}$$

2. Determine the region where the transistor is operating. Accordingly, compute the base current and the collector current. Let $V_{CC} = 10V$, $V_{BB} = 5V$, $R_C = 12k\Omega$, $R_B = 50k\Omega$, $R_E = 5k\Omega$. Assuming $\beta = 60$, $V_{BE} = 0.7\ V$ and V_{CE} at saturation is 0.1 V

   Solution

   Assume the transistor is in the active region.

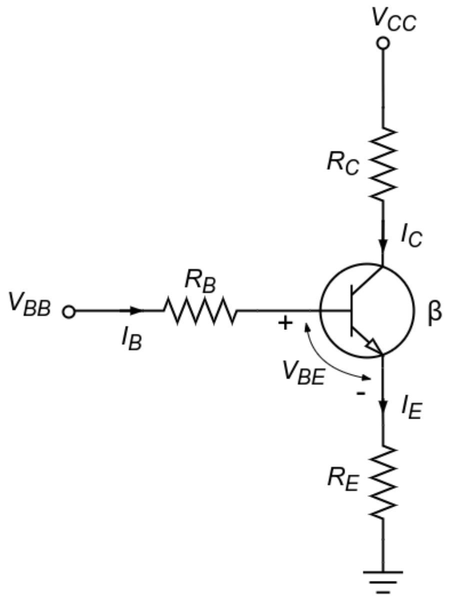

Figure 2

$$
\begin{aligned}
I_B &= \frac{V_B - V_{BE}}{R_B + (\beta+1)R_E} \\
&= \frac{5 - 0.7}{50 + 5(61)} \\
&= 12.11\mu A
\end{aligned}
$$

$$
\begin{aligned}
I_C &= \beta I_B \\
&= 60 * 12.11 \\
&= 0.726mA
\end{aligned}
$$

$$
\begin{aligned}
V_C &= V_{CC} - I_C R_C \\
&= 10 - 0.726 * 12 \\
&= 1.28V
\end{aligned}
$$

$$
\begin{aligned}
V_B &= V_{BB} - I_B R_B \\
&= 5 - 12.11 * 50 \\
&= 5 - 0.605 \\
&= 4.39V
\end{aligned}
$$

Since $V_{CB} < 0$, the transistor is not in the active region. Instead, it is in the saturation region.
Base-Emitter Loop

$$
\begin{aligned}
5 &= 50I_B + 0.7 + 5(I_B + I_C) \\
55I_B + 5I_C &= 4.3
\end{aligned}
$$

Collector-Emitter Loop

$$
\begin{aligned}
10 &= 12I_C + 0.1 + 5(I_C + I_B) \\
5I_B + 17I_C &= 9.9
\end{aligned}
$$

Solving for I_B and I_C , we get

$$
\begin{aligned}
I_B &= 25.93\,\mu A \\
I_C &= 0.57\,mA
\end{aligned}
$$

3. Calculate the line currents in the three-wire $Y - Y$ system of Figure 3.

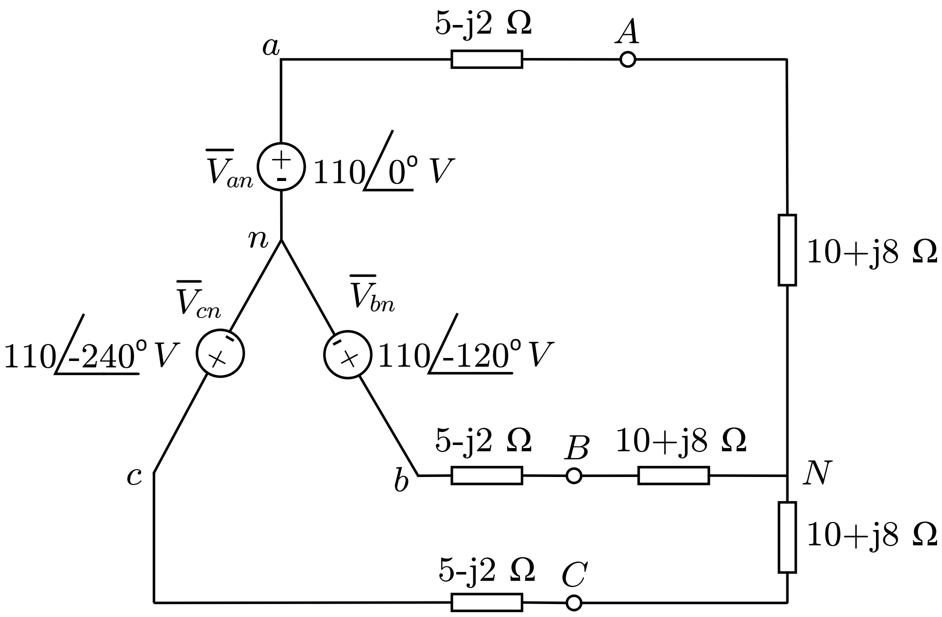

Figure 3 Three-wire $Y - Y$ system.

Solution

Note: $\overline{bar}$ indicates phasor quantity.

The three-phase circuit in Figure 3 is balanced. We may replace it with its single-phase equivalent circuit.

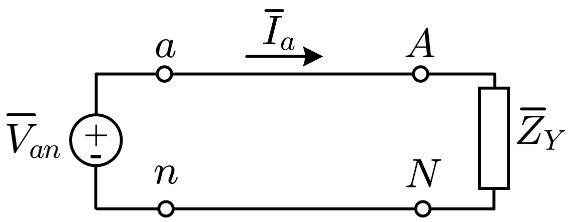

Figure 4 Single-phase equivalent circuit.

We obtain $\overline{I}_a$ from the single-phase analysis as

$$\overline{I}_a = \frac{\overline{V}_{an}}{\overline{Z}_Y}$$

where, $\overline{Z}_Y = (5 - j2) + (10 + j8) = 15 + j6 = 16.155\underline{/21.8°}$. Hence,

$$\overline{I}_a = \frac{110\underline{/0°}}{16.155\underline{/21.8°}} = 6.81\underline{/-21.8°}\ A$$

Since the source voltages in Figure 3 are in a positive sequence, the line currents are also in a positive sequence.

$$\overline{I}_b = \overline{I}_a\underline{/-120°} = 6.81\underline{/-141.8°}\ A$$
$$\overline{I}_c = \overline{I}_a\underline{/-240°} = 6.81\underline{/-261.8°} = 6.81\underline{/98.2°}\ A$$

4. A balanced abc-sequence Y-connected source with $\overline{V}_{an} = 100\underline{/10^\circ}\ V$ is connected to a Δ-connected balanced load $(8 + j4\ )\ \Omega$ per phase as shown in Figure 5. Calculate the phase and line currents.

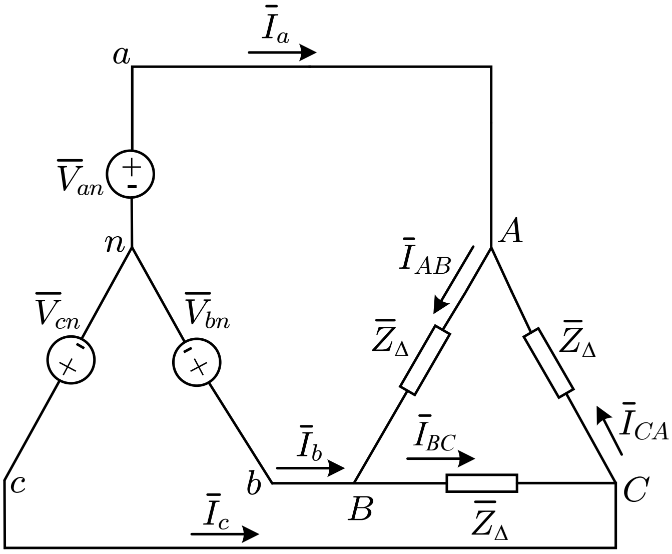

Figure 5 Three-wire $Y - \Delta$ system.

Solution

METHOD 1:

The load impedance is

$$\overline{Z}_\Delta = 8 + j4 = 8.944\underline{/26.57^\circ}\ \Omega$$

The phase voltage $\overline{V}_{an} = 100\underline{/10^\circ}$, then the line voltage is

$$\overline{V}_{ab} = \sqrt{3}\overline{V}_{an}\underline{/30^\circ} = 100\sqrt{3}\underline{/10^\circ + 30^\circ} = \overline{V}_{AB}$$

or,

$$\overline{V}_{AB} = 173.2\underline{/40^\circ}$$

The phase currents are

$$\overline{I}_{AB} = \frac{\overline{V}_{AB}}{\overline{Z}_\Delta} = \frac{173.2\underline{/40^\circ}}{8.944\underline{/26.57^\circ}} = 19.36\underline{/13.43^\circ}\ A$$
$$\overline{I}_{BC} = \overline{I}_{AB}\underline{/-120^\circ} = 19.36\underline{/-106.57^\circ}\ A$$
$$\overline{I}_{CA} = \overline{I}_{AB}\underline{/+120^\circ} = 19.36\underline{/133.43^\circ}\ A$$

The line currents are

$$\overline{I}_a = \sqrt{3}\overline{I}_{AB}\underline{/-30^\circ} = \sqrt{3}(19.36)\underline{/13.43^\circ - 30^\circ} = 33.53\underline{/-16.57^\circ}\ A$$
$$\overline{I}_b = \overline{I}_a\underline{/-120^\circ} = 33.53\underline{/-136.57^\circ}\ A$$
$$\overline{I}_c = \overline{I}_a\underline{/+120^\circ} = 33.53\underline{/103.43^\circ}\ A$$

METHOD 2:

Alternatively, using single-phase analysis,

$$\overline{I}_a = \frac{\overline{V}_{an}}{\frac{\overline{Z}_\Delta}{3}} = \frac{100\underline{/10^\circ}}{2.981\underline{/26.57^\circ}} = 33.54\underline{/-16.57^\circ}\ A$$

Other line currents are obtained using the abc phase sequence as above.

5. A balanced three-phase system with a line voltage of $300\ V$ supplies a balanced Y-connected load with $1200\ W$ at a leading PF of 0.8. Find the line current and the per-phase load impedance. Further, a balanced $600\ W$ lighting load is added (in parallel) to the system below. Determine the new line current.

Solution

Part A:

The phase voltage is $\frac{300}{\sqrt{3}}\ V$ and the per-phase power is $\frac{1200}{3} = 400\ W$. Thus, the line current may be found from the power relationship.

$$400 = \frac{300}{\sqrt{3}}(|\overline{I}_L|)(0.8)$$

The line current is, therefore, $2.89\ A$. The phase impedance magnitude is given by

$$|\overline{Z}_p| = \frac{|V_p|}{|I_L|} = \frac{\frac{300}{\sqrt{3}}}{2.89} = 60\,\Omega$$

Since the PF is 0.8 leading, the impedance phase angle is $\underline{/-36.9^\circ}$. So, $\overline{Z}p = 60\underline{/-36.9^\circ}\ \Omega$.

Part B:

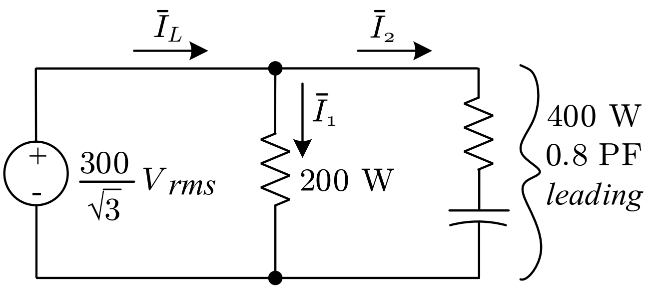

Figure 6 The per-phase equivalent circuit.

The amplitude of the lighting current (labeled $\overline{I}_1$) is determined by

$$200 = \frac{300}{\sqrt{3}}|\overline{I}_1|\cos 0^\circ$$

or,

$$|\overline{I}_1| = 1.155\ A$$

In a similar way, the amplitude of the capacitive load current (labeled $\overline{I}_2$) is found to be unchanged from its previous value since the voltage across it has remained the same.

$$|\overline{I}_2| = 2.89\ A$$

If we assume that the phase with which we are working has a phase voltage with an angle of 0°, then since $\cos^{-1}(0.8) = \underline{/36.9^\circ}$,

$$\overline{I}_1 = 1.155\underline{/0^\circ}\ A$$
$$\overline{I}_2 = 2.89\underline{/+36.9^\circ}\ A$$

and the line current is

$$\overline{I}_L = \overline{I}_1 + \overline{I}_2 = 3.87\underline{/+26.6^\circ}\ A$$

That's it.